UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

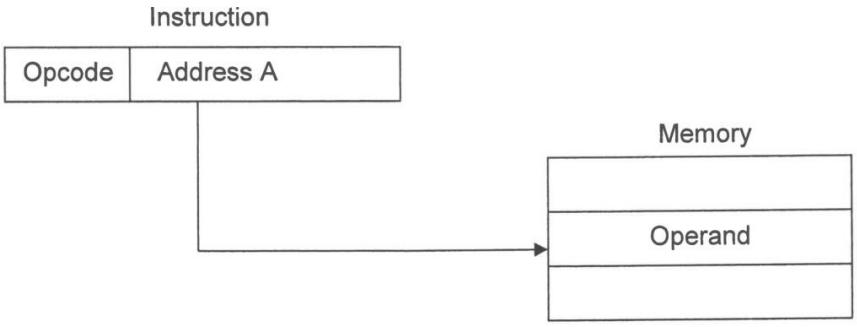
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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

Given the following Figure 2.



a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

QUESTION A2 **(5 marks)**

Complete the following taxonomy of parallel processor architecture with appropriate example.

Single Instruction Multiple Data Stream ii Processor Organization Multiple Instruction Multiple Data Stream

v

QUESTION A3 **(5 marks)**

Convert -6.75 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

QUESTION A4 **(5 marks)**

a) Discuss TWO (2) major technologies in the fourth computer generation and their impact on the modern computer.

b) Give ONE (1) example of using artificial intelligence (A1) in everyday life.

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QUESTION A5 **(10 marks)**

Given that the size of the data bus is 32 bits, the size of the address bus is 8 bits, and the size of the control bus is 4 bits.

a) Determine the size of the word and provide relevant justification.

b) Determine the size of the register and provide relevant justification.

c) Discuss what will happen if system interconnection is blocked from the perspective of the machine cycle.

d) Elaborate the role of CPU interconnection.

QUESTION A6 **(10 marks)**

a) State TWO (2) different types of bus.

b) The size of a word is 32 bits and the size of the block is 256 bits. Determine of the following:

i. The byte for word

ii. The byte for block,

iii. Size of the register,

iv. Size of the data bus.

QUESTION A7 **(10 marks)**

a) Discuss THREE (3) types of interfaces to extemal devices.

b) Classify the following input/output devices to their respective 1/0 module:

i. Network interface card

ii. Scanner Touch Screen

i. Register

QUESTION A8 **(10 marks)**

a) Explain the problem in polling 1/0 method.

b) Explain how buffering can help to improve the device overall performance?

c) Discuss how DMA technique can help the 1/0 transfer performance instead of using polling method and interrupt 1/0 method?

QUESTION A9 **(10 marks)**

a) Explain how the bits are stored in the SRAM and DFiAM.

b) Why is SRAM technology faster than DRAM technology?

c) Compare SRAM and DRAM in terms of cost and power consumption.

QUESTION A10 **(10 marks)**

Answer both parts. Show your complete workflow.

a) Convert 137710 to hexadecimal.

b) Convert 4B816 to decimal.

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b) Convert 4B816 to decimal.

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 5 and Table 1

$$z = A + C^2 / B * D - 3A$$

Write an appropriate program using conect assembly language notations to compute the equation shown in Figure 5 using instructions for zero-address, one-address, and three- address. The available instructions are provided in Table 1. The registers labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register z.

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	SUB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	